

LECTURE 17: MATH1061/7861
TOPICS: SOLVING RECURSIVE RELATIONS
DATE: WEDNESDAY APRIL 1, 2026

Learning objectives:

- (1) Reinforce recursive definitions of sequences and sets.
- (2) Prove explicit formulas and recursive formulas for sequences.
- (3) Reinforce mathematical induction.

** Take advantage of Easter Break to
Catch up and revise.
Second half is more challenging.*

Q1:

Consider the sequence $b_0 = 0$, $b_1 = 1$,

$$b_k = \frac{1}{2}(b_{k-1} + b_{k-2} + 1) + 3(k-1) \quad \forall k \geq 2.$$

(a) Calculate b_2, b_3, b_4, b_5 .

$$\begin{aligned} b_2 &= \frac{1}{2}(b_1 + b_0 + 1) + 3(2-1) = \frac{1}{2}(1+0+1) + 3 \\ &= 1 + 3 = 4 \end{aligned}$$

$$\begin{aligned} b_3 &= \frac{1}{2}(b_2 + b_1 + 1) + 3(3-1) = \frac{1}{2}(4+1+1) + 6 \\ &= 3 + 6 = 9 \end{aligned}$$

$$\begin{aligned} b_4 &= \frac{1}{2}(b_3 + b_2 + 1) + 3(4-1) = \frac{1}{2}(9+4+1) + 9 \\ &= 7 + 9 = 16 \end{aligned}$$

$$b_5 = 25$$

(b) Guess an explicit formula for b_n .

$$\underline{\underline{b_n = n^2}}$$

(c) Prove that your explicit formula is correct.

We have already seen $b_n = n^2$ is true for $n = 0, 1, 2, 3, 4, 5$

To show it is true for all n , we can use strong induction. Or we can take a bit of a shortcut and plug in the expression $b_k = k^2$ into our recursive definition

$$\text{So } b_{k-1} = (k-1)^2, \quad b_{k-2} = (k-2)^2$$

$$\begin{aligned} \& \quad b_k &= \frac{1}{2} (b_{k-1} + b_{k-2} + 1) + 3(k-1) \\ &= \frac{1}{2} (\underbrace{(k-1)^2}_{\downarrow} + \underbrace{(k-2)^2}_{\uparrow} + 1) + 3(k-1) \\ &= \frac{1}{2} (k^2 - 2k + 1 + k^2 - 4k + 4 + 1) + 3(k-1) \\ &= \frac{1}{2} (2k^2 - 6k + 6) + 3k - 3 \\ &= k^2 - 3k + 3 + 3k - 3 = k^2 \end{aligned}$$

Q2:

Define a sequence $a_0, a_1, a_2, \dots$ by

$$a_0 = 2, \quad a_1 = 5, \quad \underline{a_{n+2} = 5a_{n+1} - 6a_n} \quad \forall n \geq 0.$$

(a) Show that $a_n = 2^n + 3^n$ for all $n \geq 0$.

Indeed $a_0 = 2 = 2^0 + 3^0$

$$a_1 = 5 = 2^1 + 3^1$$

Suppose $a_{n+1} = 2^{n+1} + 3^{n+1}$ and $a_n = 2^n + 3^n$

Then $a_{n+2} = 5a_{n+1} - 6a_n$

$$\begin{aligned} &= 5(2^{n+1} + 3^{n+1}) - 6(2^n + 3^n) \\ &= (5 \times 2^{n+1} - 6 \times 2^n) + (5 \times 3^{n+1} - 6 \times 3^n) \\ &= (5 \times 2^{n+1} - 3 \times 2^{n+1}) + (5 \times 3^{n+1} - 2 \times 3^{n+1}) \\ &= (2 \times 2^{n+1}) + (3 \times 3^{n+1}) \\ &= 2^{n+2} + 3^{n+2} \quad \checkmark \end{aligned}$$

(b) Challenge: How would you guess that $2^n + 3^n$ is the solution to this recursion?

↳ Where do 2 and 3 come from?

Suppose $a_n = x^n$ for some x .

$$x^{n+2} = 5x^{n+1} - 6x^n \Rightarrow x^2 = 5x - 6$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow (x-2)(x-3)$$

(c) Can you guess explicit formula for $\Rightarrow x=2$ or $x=3$

$$F_{n+1} = F_n + F_{n-1} ?$$

$$\text{Suppose } F_n = x^n \Rightarrow x^{n+1} = x^n + x^{n-1}$$

$$[\text{divide by } x^{n-1}] \Rightarrow \cancel{x^2} = x + 1$$

$$\Rightarrow x^2 - x - 1 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow F_n = \left(\frac{1+\sqrt{5}}{2} \right)^n + \left(\frac{1-\sqrt{5}}{2} \right)^n$$